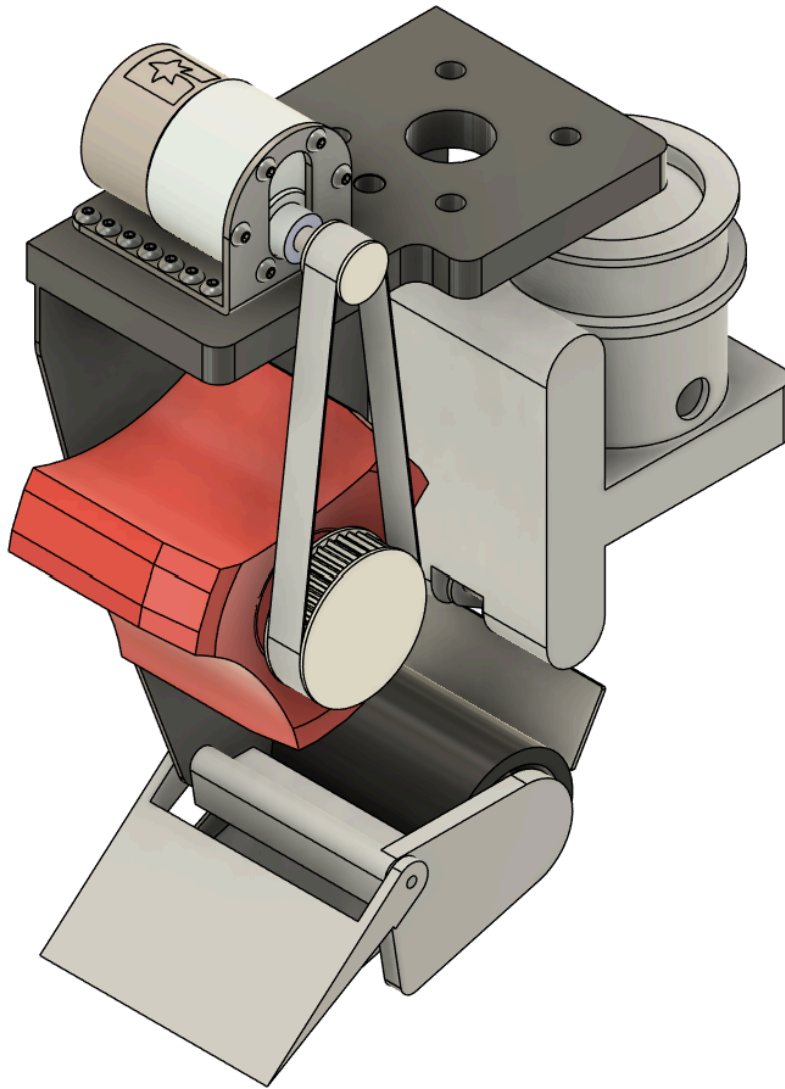


MECHANICAL DESIGN ASSIGNMENT: ROBOTIC DRYWALL TAPING TOOL



Submitted By:

Sreeram Padmesh

B.E. Mechanical Engineering

BITS Pilani, Goa Campus

Introduction

The objective of this assignment is to design a robotic drywall taping end effector that can be mounted on a standard 6-axis robotic arm to automate the process of applying joint tape and compound on drywall seams. The design has been made to achieve a simple tape storage and feeding mechanism, compound storage and application mechanism and removal of excess compound over the taped areas.

The CAD design above shows a conceptual design to achieving these goals, while also having taken into consideration aspects such as **Design for Manufacturability (DFM)** and **Design for Assembly (DFA)**. Below we will talk about the core mechanism, structural design, robot interface, user interaction, failure modes and effects analysis, manufacturing and cost, maintenance and operation.

Core Mechanism and User Interaction

There are three main mechanisms that must be understood:

- 1) Holding and feeding the paper tape
- 2) Storing and dispensing drywall compound
- 3) Applying the tape and compound to the wall with a consistent finish

Holding and Feeding the Paper Tape:

The main storage and dispensing mechanism has been inspired by packing tape gun dispensers which are usually used for taping cardboard boxes. While the product did make the taping process easier, it was not a fully automated process. By adding a DC motor and a control system, the process of taping can be automated with precise application.

To store the tape, a person must remove the pulley system, place the tape onto the tape holder (tape rolled in anticlockwise direction) and attach the pulley system back to its place. The tape must be opened up, then pulled towards the tape guider, where the user must press at the top of the tape guider, which will give way for the tape to move across but not let it wobble. The guider will go back in place when the user stops pressing as it is a spring system. Finally the tape has to be cut with the blade where the tape will rest. For a visual understanding you may refer to this video: <https://www.youtube.com/watch?v=aQFymMqDAqE>.

The feeding mechanism is done when the belt drive system comes into effect. A motor which is attached to the tape via the belt drive system starts rotating, which then rotates the tape and this allows smooth travelling of the tape through the tape guide and roller. When tape requirement is done, the robotic arm just has to slightly twist and the blade will cut the tape when required.

Storing and Dispensing Drywall Compound:

The storing and dispensing of drywall compound is made with a simple cylindrical container, a cap for the reservoir and a pipe with a nozzle which is led towards the compound guider.

The container is made for ease of use, if more compound is required, simply open the cap and add more of the drywall compound into the reservoir. This reservoir has one pipe attached firmly with a nozzle at the other end. The pipe is led to the compound guider which is a simple tool where the compound is first received. Then, while flowing down the compound will spread, and finally onto the drywall seams.

Consistent finish is achieved after the tape and compound has been applied, by a scraper that follows and removes the excess compound.

Structural Design

The design has many important components, they are listed below:

- Tape Holder
- 37D Gear Motor
- Compound Reservoir
- Chassis
- Belt Drive System
- Tape Roller
- Tape Guider
- Compound Guider
- Scraper
- Pipe

Below is a table in which the materials used are given and why these materials were chosen:

Component Name	Material Used	Explanation
Tape Holder	High Density Polyethylene	HDPE has low friction and toughness which prevents tape damage, additionally being corrosion proof and lightweight
37D Gear Motor	Store Bought	-
Compound Reservoir	High Density Polyethylene	HDPE is cheap, light and easy to clean while also not allowing the compound to stick easily
Chassis	Steel ASTM A36	Excellent machinability, good yield strength and dimensional stability under load
Belt Drive System	Nylon - Pulley PVC - Belt	Nylon - Low mass, low noise, lubricous and corrosion proof PVC - Grippy, good traction on nylon and easy to clean
Tape Roller	Acetal	Low friction, hard and slick surface with excellent machinability
Tape Guider	Stainless Steel	Corrosion resistant, hard and smooth surface
Compound Guider	Stainless Steel	Smooth, hard and cleanable surface, allowing compound movement, while also being rust resistant
Scraper	Stainless Steel	Rust resistant to compound material and smooth, hard and cleanable surface
Pipe	Convolutd PTFE	very flexible, chemically inert, easy to clean; light

Robotic Interface

The mounting can be done at the top, near the motor where the right holes have been made for ease of mounting. The robotic arm is assumed to have a male spigot of 20mm and hence there is 20mm hole at the center. Align the orientation such that the 6mm hole is in line

with the robot pin. Then bolt up the 4 M6 clearance holes by inserting M6 screws from underneath. Finally verify and check proper mounting.

Failure Modes and Effects Analysis

1) Joint slip due to heavy load concentration:

- a) Potential Effect - Yielding of the bolt, clamping force holding the joint reduces drastically, loosening the joint and further increasing vibration and movement, causing it to handle cyclic stresses
- b) Potential Cause - Joint holding the reservoir may have this problem due to the heavy load of the reservoir and addition weight of the compound itself
- c) Proposed Test - A static tensile load testing in which the load is increased till the bolt yields or fractures. A fatigue test where the joint is subjected to cyclic loads and checked for the number of cycles to failure

2) Tape tears

- a) Potential Effect - Application of tape may be halted, manual interaction may be required to remove excess/wasted tape and restart the process
- b) Potential Cause - When tape feeding speed does not match the speed at which the robotic arm is moving. As the tape radius varies, the speed at which the tape will be fed will vary.
- c) Proposed Test - Use an encoder and test the linear speed at which the tape is being fed and write a control algorithm for the linear speed of the entire end effector

3) Belt slip or break

- a) Potential Effect - Under-feed or velocity ripple causing poor finish; sudden stop if belt breaks; possible tape tear; operator must re-thread/replace belt
- b) Potential Cause - Low pretension or warp angle, belt contaminated with compound, worn belt, misaligned belt
- c) Proposed Test - Test the belt with varying pretensions and check when the tape is fed properly without tear or ripples and do proper cleaning and alignment always while testing or using the robot

4) Scraper edge wear

- a) Potential Effect - Wearing and dents on the scraper edge requiring us to change the component
- b) Potential Cause - The edge of the scraper may wear from abrasive dust in the compound and dents may be caused if it comes in contact with a screw or other protruding components on the wall
- c) Proposed Test - Abrasion endurance test, mount the scraper at an angle and test it with drywall compound with a controlled stroke and figure out the edge recession after 10000-20000 strokes

Manufacturing and Cost

There are many components that are part of this end effector and the list of the primary manufacturing processes used are mentioned below, giving context about the parts using these specified methods.

1) Laser Cutting:

Fast and accurate flat profiles while being cheap and repeatable. Used for the **main chassis** 1.5mm thickness flat plate with unique cross section, **tape guide**, **compound guide** and **scraper**..

2) VARTM (Vacuum-Assisted Resin Transfer Molding):

Gives a thick, stiff GFRP mold in one shot with low voids and consistent thickness. Used for the **reservoir holder** which looks like a T shape.

3) CNC Machining:

precise, fast for low-to-mid volumes; avoids sink/warpage. Used for making the **tape holder**. CNC milling will be used for stainless steel components such as **tape guide** and **compound guide**. CNC turning is used for making the **tape roller** made from acetal.

4) TIG Welding:

Used for joining different metallic components of the **tape guide**

5) Drilling:

Used for drilling holes in which the mounting flange of the robotic arm will attach to the end effector

Cost: Rs 35000 - 50000

Lead Time: 10-15 days

Maintenance and Operation

Cleaning Protocol (end of day)

1. Power down
2. Lift the scraper and wipe off wet compound, inspect edge, then rinse and dry
3. At the compound guider, scrape the residue, rinse, and dry and check that the guide face is smooth
4. Open reservoir cap; drain any leftover slurry, clean the reservoir with water, then wipe it and dry it open
5. Wipe belt, pulleys, tape guider faces

Setup Procedure (start of job)

1. Mount tool to robot flange and verify the movement of the end effector
2. Remove the belt-drive system, both the pulleys and the belt
3. Load tape: click the tape roll onto the HDPE holder (roll wound anticlockwise), refit pulley system
4. Thread tape: pull free end to the tape guider, then press the top of the spring-loaded guider to open the path and feed the tape through and release after so it self-clamps and prevents wobble, cut at the resting blade
5. Place the belt drive system back
6. Fill compound: open reservoir cap, pour premixed compound, close
7. Run a dry-run on a scrap board

Ergonomics & Safety (design features and practice)

- Belt & pulleys: interact only when power is down and keep fingers clear of pinch points when tensioning
- Blade/cutter: carefully clean the tape guide and scraper near the blade
- Compound handling: use non-chloride cleaners; wipe spills immediately; keep reservoir cap tethered to avoid drops

Payload

The payload capacity of the end effector is 8kgs and the weight of the design I have made is equal to weight of fusion design + weight of pipe + weight of compound when fully loaded + weight of full tape.

Total weight = $1911\text{g} + 15\text{g} + 200\text{g} + 650\text{g} = 2776\text{g}$, which is much lesser than the 8kg payload capacity. But, the reservoir volume is too low that only 0.2kg of the compound can fit in the reservoir. One method is that, there is one more reservoir which is carried by the robot, that can be used to take more compound into the reservoir at the end effector or just increasing the volume of the cylinder. Precaution was taken on the weight of the reservoir with its compound as the holder was bolted to a joint.

Conclusion

The design can have a significant number of improvements, such as the weight of compound the reservoir can hold and more payload utilization. But the design has a lot of promise and potential to work as the taping system that was inspired by the packaging tape gun dispenser is a very simple but reusable idea in this case.

The assignment taught me to research working mechanisms before going straight into designing and understanding the purpose of the product that I am working on. I would love to hear some feedback on how I have presented my work and about the work itself.